

Math 243: Homework 02

Due Wednesday, September 9

Problem 1. Consider the parametric equations

$$\begin{cases} x = 2t - 5 \\ y = 4t - 7 \end{cases}, \quad -\infty < t < \infty.$$

(a) Identify the ant's path by finding a Cartesian equation for it.

(b) The **vector equation of a line** is

$$\mathbf{r}(t) = \mathbf{v}t + \mathbf{b}, \quad -\infty < t < \infty$$

where

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}.$$

(Note the similarity to the familiar line equation $y = mx + b$.)

Write the ant's path in this form, and use it to identify the direction of the ant's travel, and its position at $t = 0$.

Problem 2. An ant walks around the ellipse

$$\mathbf{r}(t) = (4 \cos t + 3, 2 \sin t + 1), \quad 0 \leq t \leq 2\pi.$$

Compare the ant's speed at $t = 0$ with its speed at $t = \pi/2$. Does the ant travel around the ellipse at constant speed?

Problem 3. Two ants walk the same circular track. Ant A has $\mathbf{r}_A(t) = (\cos t, \sin t)$ and Ant B has $\mathbf{r}_B(t) = (\cos 2t, \sin 2t)$. Compare their velocity vectors, speeds, and directions at $t = \pi/4$, by filling out the following table. Which of these three quantities agree?

	Velocity	Speed	Direction
Ant A			
Ant B			

Problem 4.

(a) Give an example of a parametric curve that cannot be written as the form in Cartesian form $y = f(x)$.

(b) **(True/False)** Every curve that can be written in the form $y = f(x)$ can also be described by parametric equations. Justify your answer.

(c) Theorem 1 from the lecture notes says that the graph of $y = f(x)$ over $[a, b]$ has arc length

$$(\text{arc length}) = \int_a^b \sqrt{1 + [f'(x)]^2} dx.$$

Show that this formula is implied by Theorem 18 from the lecture notes. (Hint: consider the parametrization $\mathbf{r}(t) = (t, f(t))$, $a \leq t \leq b$. [This shows that Theorem 1 is just a special case of Theorem 18].

Problem 5. In Example 15 of the lectures, we eliminated the parameter from

$$\begin{cases} x = 5 \cos^2(\theta) \\ y = 2 \sin^2(\theta) \end{cases}$$

and obtained the line $y = -\frac{2}{5}x + 2$. Explain in a sentence or two *what information was lost* in that computation, and why the restriction $0 \leq x \leq 5$ had to be added by hand.

Problem 6. Explain why the ellipse equation $\left(\frac{x-h}{a}\right)^2 + \left(\frac{y-k}{b}\right)^2 = 1$ cannot, by itself, tell you the direction or speed of a particle traveling around the ellipse, whereas the parametric form can.

Problem 7. Suppose we have a parameterized curve $\mathbf{r}(t) = (x(t), y(t))$ such that at some time t_0 we have $x'(t_0) = 0$ but $y'(t_0) \neq 0$. What does this tell you about the ant's motion at that instant, and what does it tell you about the shape of the curve there?

Problem 8. The distance traveled by an ant along a curve $\mathbf{r}(t)$, $a \leq t \leq b$ is

$$(\text{distance traveled}) = \int_a^b |\mathbf{r}'(t)| dt$$

Why do we integrate $|\mathbf{r}'(t)|$ rather than $\mathbf{r}(t)$? What would $\int_a^b \mathbf{r}'(t) dt$ compute instead, and why is that a different quantity?

Problem 9. During the short time interval $[t, t + \Delta t]$, we approximate the ant's distance traveled by

$$(\text{distance traveled}) \approx |\mathbf{r}'(t)| \Delta t$$

This formula pretends that the ant travels at the constant speed $|\mathbf{r}'(t)|$ throughout the interval, it's generally not the exact distance traveled. Explain why this approximation improves as the interval gets shorter.